

ENM 5310 — Quiz 2: Solutions

Lectures 4–7 · 6 questions, 1 point each

Your quiz sheet was Version A, B or C — the questions are identical and in the same order, but the answer options are permuted. Find your version's column below.

Q	Topic	Version A	Version B	Version C
1	What maximum likelihood converges to	C	B	D
2	Least squares: geometry and the noise model	A	D	B
3	Ridge as MAP	B	C	A
4	Gradient descent vs. Newton's method	C	A	B
5	Logistic regression: gradient and convexity	D	B	C
6	Aleatoric vs. epistemic uncertainty	B	D	A

1. What maximum likelihood converges to. *Correct: it minimizes $\text{KL}(p^*||p_\theta)$ and straddles both modes.*

Maximum likelihood minimizes the cross-entropy between the empirical distribution and the model; as $N \rightarrow \infty$ the objective converges to $\mathbb{H}(p^*, p_\theta) = \text{KL}(p^*||p_\theta) + \mathbb{H}[p^*]$, whose second term is free of θ . So $\hat{\theta} \rightarrow \arg \min_\theta \text{KL}(p^*||p_\theta)$. That divergence is an average *weighted by p^** : it charges $+\infty$ wherever the data has mass and the model has none, and charges nothing for mass invented where there is no data. The fit is therefore *mass-covering* — a wide Gaussian bridging both modes, placing most of its probability exactly where nothing was observed. This is not an optimizer failure; it is the exact minimizer of the divergence we chose. (Lecture 4, §2.)

The tempting wrong answer is the reverse divergence $\text{KL}(p_\theta||p^*)$, which *is* mode-seeking — but that is the objective of variational inference, not of maximum likelihood. Which direction of KL a method minimizes determines which way it misleads you.

2. Least squares: geometry and the noise model. *Correct: orthogonal projection, and exactly the MLE.*

Setting $\nabla_w \|y - \Phi w\|^2 = -2\Phi^\top(y - \Phi w)$ to zero gives the normal equations $\Phi^\top(y - \Phi \hat{w}) = 0$: the residual is orthogonal to every column of Φ . Equivalently $\hat{y} = \Phi(\Phi^\top\Phi)^{-1}\Phi^\top y = Hy$ with H symmetric and idempotent — the defining properties of an orthogonal projection onto the column space of Φ . Least squares is a projection; everything else is bookkeeping.

It is also *exactly* maximum likelihood, not an approximation to it at any N . The Gaussian NLL is $\frac{N}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \|y - \Phi w\|^2$, in which σ^2 multiplies the w -dependent term by a positive constant — so the minimizing $\hat{w}$ is the same for every σ^2 . The important direction of the argument is that **the noise model determines the loss**, not the reverse: a Laplace noise model would have produced ℓ_1 loss and a robust fit, a Student- t model something more robust still, and per-point variances σ_n^2 would have produced weighted least squares. Squared error is a modeling commitment, not a convenience.

Finally, “linear” refers to w , not to x . With ϕ any fixed map — polynomials, Fourier features, radial basis functions — the model is nonlinear in x while every formula above is unchanged. That is the hinge on which kernel methods will turn. (Lecture 3, §4; Lecture 4, §6.)

3. Ridge as MAP. *Correct: $\lambda = \sigma^2/\tau^2$, and the estimate always exists.*

$\arg \max_w [\log p(y | w) + \log p(w)]$ gives $\frac{1}{2\sigma^2} \|y - \Phi w\|^2 + \frac{1}{2\tau^2} \|w\|^2$, i.e. $\|y - \Phi w\|^2 + \lambda \|w\|^2$ with $\lambda = \sigma^2/\tau^2$: regularize in proportion to the noise, in inverse proportion to the expected coefficient size. λ is a ratio of variances, not an arbitrary knob. Adding λI adds λ to every eigenvalue of the positive semi-definite $\Phi^\top\Phi$, so the solution exists and is unique for any $\lambda > 0$ — the prior supplies curvature exactly where the likelihood was flat. Ridge is *biased* (toward zero, most strongly in the poorly-determined directions); it buys a larger reduction in variance with that bias. (Lecture 5, §2.)

4. Gradient descent vs. Newton's method. *Correct: Newton minimizes the local quadratic model, at $O(D^3)$ per step.*

Gradient descent builds a *linear* model of f , and a linear model has no minimum — so a trust radius has to be imposed from outside, and it survives in the algorithm as the step size η . Newton's method builds a model that does have a minimum: truncate the Taylor expansion at second order,

$$m(\delta) = f(\theta) + \nabla f(\theta)^\top \delta + \frac{1}{2} \delta^\top \nabla^2 f(\theta) \delta,$$

and minimize it exactly, $\nabla_\delta m = \nabla f + \nabla^2 f \delta = 0$, giving $\nabla^2 f(\theta) \delta = -\nabla f(\theta)$. Two consequences follow directly. No step size is invented, because the length of the step comes out of the curvature. And if f really is quadratic the model is exact, so a single step lands on the minimizer from any starting point. (Lecture 6.)

Three things the wrong options get wrong. The Newton *direction* is $-(\nabla^2 f)^{-1} \nabla f$, which rescales each direction of parameter space by its own curvature — long steps where the objective is flat, short where it is steep — so it is generally *not* parallel to $-\nabla f$. Newton needs only that f be twice differentiable, not quadratic; the quadratic is simply the case where the model is exact. And both methods converge to stationary points, so curvature buys speed, not a better minimum.

The real cost is arithmetic: forming and factoring the Hessian is $O(D^3)$ per iteration with $O(D^2)$ memory, which is fine at $D = 10$ and impossible at $D = 10^9$ — hence SGD and Adam for large models.

5. Logistic regression. *Correct: gradient $\Phi^\top(\mu - y)$, no closed form, but convex.*

Differentiating the stable form $\text{softplus}(a_n) - y_n a_n$ gives $\partial/\partial a_n = \sigma(a_n) - y_n$, so by the chain rule $\nabla_w \text{NLL} = \Phi^\top(\mu - y)$ — the same $\Phi^\top(\text{prediction} - \text{target})$ form as linear regression. The difference is that $\mu = \sigma(\Phi w)$ is *nonlinear* in w , so $\Phi^\top(\sigma(\Phi w) - y) = 0$ is transcendental and no algebraic rearrangement isolates w . This is why the course needs optimization at all. The Hessian is $\Phi^\top S \Phi$ with $S = \text{diag}(\mu_n(1 - \mu_n)) \succ 0$, so the objective is convex: no local minima, and one Newton step is *not* exact because S depends on w . (Lecture 5, §§6–7.)

6. Aleatoric vs. epistemic uncertainty. *Correct: only the second term shrinks with N .*

σ^2 is the assumed observation noise: it does not depend on x^* , and no amount of data reduces it. The term $\phi(x^*)^\top S_N \phi(x^*)$ is uncertainty about w pushed through to the prediction: it depends on where you ask, and since $S_N^{-1} = \tau^{-2} I + \sigma^{-2} \Phi^\top \Phi$ grows linearly in N , it shrinks like $O(1/N)$. So the predictive variance tends to σ^2 , never to zero: **with unlimited data you can learn the function exactly and still not predict the next measurement exactly.** If your predictive intervals do not show that behavior, something is wrong. Only the epistemic part is reducible by collecting data — which is why the acquisition functions of Bayesian optimization are built from it alone. (Lecture 7, §6.2.)

To review: the relevant sections are cited after each solution. Questions 1, 4, and 6 are load-bearing for the rest of the course — Q1 sets up variational inference, Q4 is the diagnosis you will run on every training curve, and Q6 is the foundation of Gaussian processes and Bayesian optimization. If any of the three is still unclear, come to office hours this week.